SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE & NAME CSA16/Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Ajay I (192421147) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AJAY I

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1. Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1. Dataset: Customer Data

Customer	**Income (₹)**	**Spending Score**
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	**Word1**	**Word2**	**Word3**
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	**Latitude**	**Longitude**
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyse their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	**Sales (₹)**	**Frequency**
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5. Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyse their effectiveness on mixed data types. *(BL4 – 1 Mark)*

Answers:

Q1. K-Means vs Hierarchical Clustering

The question compares both methods using the customer income and spending-score dataset.

Feature	K-Means Clustering	Hierarchical Clustering
Method	Divides data into a fixed number of clusters (K).	Builds a hierarchy of clusters.
Working	Assigns data points to the nearest centroid.	Merges or splits clusters based on similarity.
Number of clusters	K must be specified.	Number of clusters can be selected from the hierarchy.
Performance	Faster and efficient for large datasets.	Slower for large datasets.
Scalability	Highly scalable.	Less scalable.
For this dataset	Suitable and simple to apply.	Suitable because the dataset is small.
Conclusion	**Better for speed and scalability.**	**Better for understanding cluster relationships.**

Q2. K-Means vs EM Algorithms

The question focuses on handling **overlapping clusters** in the document dataset.

Feature	K-Means	EM (Expectation-Maximization)
Cluster assignment	Assigns each document to one cluster.	Gives probabilities of belonging to different clusters.
Overlapping clusters	Not very effective.	More effective.
Working	Uses distance from cluster centroids.	Estimates probability distributions.
Flexibility	Less flexible.	More flexible.

Complexity	Simple and fast.	More complex and computationally expensive.
For this dataset	May not handle similar documents well.	Handles overlapping document groups better.
Conclusion	Suitable for clearly separated clusters.	**Better for overlapping clusters.**

Q3. Euclidean vs Manhattan Distance

The question asks how the two distance measures affect clustering of geographic points.

Feature	Euclidean Distance	Manhattan Distance
Meaning	Measures straight-line distance between two points.	Measures distance using the sum of coordinate differences.
Formula	$\sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$	$\|x_2-x_1\| + \|y_2-y_1\|$
Path	Direct/straight path.	Grid-like path.
Sensitivity	More affected by large differences.	Less affected by individual large differences.
Effect on clustering	Groups points based on straight-line closeness.	May produce different groupings based on coordinate differences.
Geographic data	Generally useful for direct spatial distance.	Useful when movement follows grid-like paths.
Conclusion	Can produce one set of geographic clusters.	Can produce different cluster boundaries.

Q4. Agglomerative vs Divisive Clustering

The question compares the two hierarchical approaches for the retail product dataset.

Feature	Agglomerative Clustering	Divisive Clustering
Approach	Bottom-up approach.	Top-down approach.
Starting point	Each product starts as a separate cluster.	All products start in one cluster.
Process	Repeatedly merges similar clusters.	Repeatedly splits clusters.
Complexity	Generally simpler to implement.	More complex.
Dataset suitability	Very suitable for small datasets.	Can also be used for small datasets.
For retail data	Easily groups products with similar sales and frequency.	Splits products into different groups based on similarity.
Conclusion	**More suitable for this small dataset.**	Suitable but comparatively more complex.

Q5. Centroid-Based vs Density-Based Clustering

The question compares both techniques for mixed attributes: price, category and rating.

Feature	Centroid-Based Clustering	Density-Based Clustering
Example	K-Means	DBSCAN
Basic idea	Groups data around a central point (centroid).	Groups data based on dense regions.
Data requirement	Works mainly with numerical data.	Works with numerical distance-based data.
Mixed data	Category needs encoding before use.	Category also needs suitable preprocessing/encoding.
Outliers	Sensitive to outliers.	Can identify outliers as noise.
Cluster shape	Works best with compact/spherical clusters.	Can identify irregularly shaped clusters.
For this dataset	Simple after encoding category.	Useful if there are irregular groups or outliers.
Conclusion	**Simple and suitable after preprocessing.**	**Useful for noise and irregular clusters.**